Mars Telecommunications Network (MTN)
Draft Request for Proposal: 80GSFC26R0011
Questions and Answers
May 08, 2026

Question No.	RFP Document	Question	Government Response
1	MTN-402-REQ-0001	MTN-402-REQ-0001: For both DSN compatibility and Mars relay user compatibility testing, does NASA expect to provide any test assets, such as ground station/Mars relay user simulators and test equipment?	NASA intends to provide access to CTT-22, MIL-71 and DTF-21 as indicated in Section 4 Ground Rules and Assumptions. This access will have to be negotiated with NASA. NASA also intends to provide access to UHF relay compatibility test sets as identified in Attachment E Government Furnished Property (GFP), assuming this option of the GFP is chosen.
2	MTN-REQ-005	*"The contractor shall deliver completed ground and flight systems to NASA including completion of all associated reviews and inspections by December 31, 2028."* Please confirm that December 31, 2028 is the intended no later than delivery date since the late 2028 Mars launch window requires launch vehicle integration no later than July 2028 assuming a two month launch vehicle integration.	MTN-REQ-005 establishes the statutory (driven by P.L. 119-21) delivery requirement, which should be viewed as a no later than date. NASA intends to accept delivery of a data package associated with the spacecraft at a pre ship review (or vendor equivalent) prior to launch vehicle integration as the culmination of CLIN 1. MTN-REQ-013 (operational readiness at Mars by end of 2030) is driven by urgency to make the MRN more robust and is the more stringent schedule driver. Upon successful completion of checkout and commisioning at Mars, NASA intends to accept DD250 delivery of the spaceflight hardware.
3	MTN-REQ-006	Is the DWE link for telecommunications relay meant to be 1-way or 2-way?	The DWE link is required to be two-way as defined in MTN-REQ-024 and MTN-REQ-041. That two-way communication can be either (FWD/RTN) X/X or X/Ka. If the contractor chooses to implement Ka forward then that is a third configuration. However, Ka forward is not a threshold requirement.
4	MTN-REQ-006	Also, if this is a 2-way link, what is the motivation behind MTN-REQ-042, which seeks an autonomous navigation solution (given that a 2-way link provides ground-based Doppler for free with the DWE link)?	The 2-way link does provide ground-based radiometrics that can be used in ground-based orbit determination and uplinked to the flight system(s). MTN-REQ-042 establishes the goal of autonomous navigation with the objective of reducing load on the ground elements (e.g., DSN) and ground personnel. It seeks to reduce the dependence of the MTN flight system on Earth-based resources to enable system level autonomy (e.g., near real time relay schedule updates, antenna pointing updates, etc.)
5	MTN-REQ-013	*"Mars Telecommunications Network shall achieve full operational readiness at Mars no later than end of calendar year 2030."* In combination with the industry slide, this seems to give room for EP cruise with longer durations. For EP spiral to lower Mars orbits, there can be a range of transfer windows depending on the desired end altitude. Additionally, with the delta-v provided by an EP system, the spacecraft could potentially be re-positioned between different operational orbits. Based on this, could you clarify the definition of full operational readiness? In the case that re-positioning the orbiter is deemed advantageous, and communications can be supported during transfer windows, would this re-position period still count towards "full operational readiness?"	The intent of "full operational readiness" is that spacecraft checkout and commissioning are complete at Mars and that the system is ready to begin nominal operations at the planned end-state orbit. Additonal delta-v for adjusting the orbit (e.g., as may be necessary for an EDL event) is deemed advantageous for the system's extensibility.

6	MTN-REQ-019	*"The Mars Telecommunications Network shall have the onboard capability to support continuous and full rate data transfer between the Direct with Earth links and the Proximity links, and between different Proximity links."* Please confirm that *"continuous and full data transfer"* between links applies only when the links are active. Otherwise, the requirement implies that the link systems need to be continuously active even though proximity links with users may not be established or active.	The full-rate data transfer between the links is only when the links are active. For example, one UHF link (lander) could transfer data to another UHF link (lander) through the link. Similarly, one Advanced Proximity link could be expected to transfer data to the DWE link.
7	MTN-REQ-022 (Ref. MTN-REQ-039)	*"Mars Telecommunications Network shall be able to support 2-way coherent ranging and 2-way coherent Doppler on the DWE links with a range precision 1m and Doppler precision of 0.1 mm/s (1-sigma) at a 60 second count time."* Please clarify "precision" and if precision is an accuracy metric in this context. A proximity link pseudorange accuracy of 1m is very challenging when accounting for clock error (see MTN-REQ-039 where the allowed Mars SISE is 100 m (T)/50 m (G); X-band DSN DWE links are typically 3-5 m (1-sigma) in range error and even larger when incorporating per-pass range biases).	This requirement refers to the two-way range and Doppler used to determine the MTN orbit and specifies the measurement precision of the MTN to Earth link. In the requirement, precision refers to the measurement noise induced by link budget considerations and noise from the MTN DWE transponder. The 1m (1 sigma) range precision value is readily achievable with sufficient antenna gain as would be expected for the high-rate link that the MTN will have with the Earth (for reference see DSN 810-005 203 Figure 14 that is useful for a higher-powered links). The requirement does not include effects that limit accuracy such as per pass range bias errors (which current practice is on the order of 2-3 meters 1 sigma), media delay effects, Earth orientation errors, etc… This requirement does not represent values to be satisfied by the proximity link.
8	MTN-REQ-030	*"Mars Telecommunications Network shall provide an average annual Proximity Link service availability ≥98% during the Proximity Communications windows."* Does the 1-way PNT signal need to be available at all times even when maneuvering to different orbits?	The signal shall be available when the proximity link is active but can be considered to be degraded (i.e., similar to a GPS unhealthy flag) over the period of time a maneuver has degraded the broadcast signals accuracy. However, the MTN shall reconstruct orbits and provide that reconstruction to users for post-processing.
9		In order to confirm that the contractor and NASA are using the same definitions and assumptions, for all doppler, ranging, and pseudoranging accuracy, precision, and error requirements, could NASA provide a sample calculation or error budget template, similar to what was provided for the link budget templates?	Additional guidance on computing and evaluating quality of MTN orbit determination and Mars time determination and how it relates to the SISE is being provided as a narrative that will appear as an appendix in the requirements document.
10	MTN-REQ-035 (ref. MTN-REQ-037)	*"The Mars Telecommunications Network shall provide in situ 1-way Doppler tracking services to users with a precision of 0.5mm/s (1 sigma) at a 60 second count time."* Please describe the 1-way Doppler tracking service intent and CONOPS. Is 1-way Doppler only useful for end users with very good clocks (i.e.: DSAC)? Can existing user and future user clock/oscillator performance be provided? Should 1-way Doppler be considered unusable until other satellites are added to the MTN constellation? The MTN-REQ-037 threshold requirement includes both pseudorange and 1-way Doppler, yet MTN-REQ-035 threshold requirement includes 1-way Doppler only. Should we assume pseudorange is a threshold requirement as well since MTN-REQ-035 and MTN-REQ-037 are related?	The one-way Doppler signal is still useful for users that have oscillators with poor stability as long as their positioning or orbit determination needs are modest. For instance, the current MRN provides 1-way Doppler tracking to the Mars rovers (i.e., M2020 or MSL) that can be used for localization and has been shown to recover 2 km class position knowledge when the oscillators at either end of the link are only TCXOs (as is the case for the rovers and Maven and TGO). The requirement is for the Doppler noise induced by the MTN relay and includes typical thermal errors (assume a 0 db gain user antenna) and the relays source oscillator noise (which will dominate). The 0.5 mm/sec at 60 seconds is equivalent to an Allan Deviation of 2e-12 at 60 seconds, which is readily achievable with available ovenized space qualified oscillators or space qualified miniature rubidium atomic clocks (data sheet provided on request) that will be available beginning Fall 2026. The Doppler precision is simply a precision and not accuracy, the longer-term effects of the MTN local oscillator will manifest in the SISE. Choosing a more stable local oscillator will increase the period of time that the MTN will need to update its broadcast navigation message.

11	MTN-REQ-036	*"The Mars Telecommunications Network shall provide for in situ 2-Way coherent Doppler tracking services to users with a precision of 0.5mm/s (1-sigma) at a 60 second count time with a latency of less than 10 seconds."* Is it NASA's intention that the 2-way PNT solution needs to be computed on-board MTN and immediately transferred to the end user? Or, can the 2-way Doppler/Ranging measurement be transferred to the Ground for processing and transferred back to the end user once the Ground processing result is received by the MTN orbiter?	The intent here is to have the ability to collect the 2-way range/Doppler by the MTN (with the user transponding/transceiving) and timestamp the data using the MTN representation of Mars time (traceable to UTC) and then telemeter to the user on the MTN forward link as a data message to the user (that is also likely still being used for the radiometric tracking). This supports the class of user that has a realtime PNT need. This data should also be sent back to Earth as part of the user's data transmission along with other data being transmitted by the user for more traditional ground-based PNT processing. The direction of the data transfer would be an option the user can select.
12	MTN-REQ-042	*"MTN shall estimate its navigation state onboard without utilization of Earth- based radiometric tracking."* Is NASA's expectation that autonomous navigation is available immediately, or can the feature be added later in the mission life?	The hardware and compute capacity must be verifiably sufficient to meet the goal. It can be added later - such software development tasks should be included in the pricing.
13	MTN-REQ-039	Describes an open-loop recording CONOPS. Attachment G, Government Furnished Equipment (GFE) states that the available Electra radio provides *"concurrent open-loop record and closed-loop demodulation, and higher sampling rate open-loop recordings."* In lieu of more specific data sheets on Electra, can NASA confirm whether Electra meets MTN-REQ-039 as written or if a combination of Electra and other radio capabilities will be needed to meet this requirement?	The Electra Radios meet this requirement. However, this requirement is also levied on the Advanced Proximity Link. In addition, the requirement states: "at a selectable sampling rate ≤ 2 MHz ". At higher data rates on the UHF Proximity Link the sampling rate might be reduced for open loop recording. Typically this feature is used for either RF sniffing or recording a critical event (in addition to demodulation).
14	MTN-OBJ-001	*"Provide communication and data exchange between assets in the Mars vicinity, the Mars surface, and Earth anticipated to operate at Mars through 2035."* In order to provide better pricing estimates for O&M services, can NASA please identify current and future expected missions to be supported, use cases and communications requirements through 2035, and transfer these into aggregate demand volume measured in minutes of service (NASA's traditional service metric unit).	This is combined answer for all questions related to use cases that drove MTN-OBJ-001 - MTN-OBJ-004. The following are example users: Legacy surface assets (Curiosity, Perserverance) New surface assets (Rosalind Franklin, SkyFall) Small science orbiters "human scale" Entry Descent and Landing demonstrations Future looking use cases will evolve as plans and timelines mature. Communications requirements: UHF proximity link (1500 Mb/sol) + TBD SkyFall requirment; Advanced proximity link: 11.5 Gb/sol; EDL demo: 8Gb/sol for 14 sols. It is expected that small science orbiters and EDL events will leverage the PNT services specified in the requirements.
15	MTN-402-REQ-0001	1.1 Purpose states, *"The intent of these objectives and requirements is to support definition of the required "what" and to the maximum extent possible not define the "how"* ." However, there were a significant number of meaningful requirements changes between version "MTN-402-REQ-0001, Preliminary" and version "MTN-402-REQ-0001, Rev A" of the document. Including 11 new technical requirements. How does this align with that statement?	NASA provided prescription of implementation, particularly frquency selection, where it provided clarity (DSN compatibility) or we had a strong reason to prefer specific implementations (adv. Prox links). Other changes that tended to precriptiveness were meant to narrow the solution set to what NASA forsees as its needs.
16	MTN-402-REQ-0001	1.1 Purpose states, *"The longer-term vision is evolution of the existing Mars network to a higher performing system with extensibility to a 24/7 end state."* Please clarify what you mean by 24/7 end state.	The long-term vision for the Mars network is to be able to provide relay services to any assets on or near Mars at any time. This capability is not a requirement for MTN, which accepts intermittent contacts.

17	MTN-REQ-049	MTN-REQ-049 : includes a threshold requirement to accommodate a NASA science payload. Please add a goal requirement to accommodate as many NASA science payloads as can be hosted or deployed above the threshold.	To keep the procurement selection primarily focused on telecommunications and navigation system needs, NASA declines to add this goal. This does not suggest that payload can't be added subsequent to the selection, but such additions must not impact MTN core telecommunications and navigation capabilities.
18	MTN-REQ-049	Includes a threshold requirement to accommodate a NASA science payload. However, it doesn't specify the kind of science that is to be performed, and how this might drive the desired orbital altitude of MTN. For example, Mars weather observation or high-resolution imagery for EDL site selection and surface mission operations support would desire to be operated in a low altitude orbit. This approach, especially supported imagery, would support similar use cases being defined for the orbital assets at the Moon in support of recent Lunar base initiatives. Can NASA clarify the intended altitude for science payload use cases?	The science payload will not drive orbit selection, pointing, or altitude. Payload selection is not yet complete. NASA is considering deployable CubeSat options that may relieve operational considerations like pointing, etc.
19	Attachment F - Draft DRDs	Data Requirements Deliverable (DRD), articulates 54 unique DRDs. This is a significant cost and schedule driver. The Offeror recommends NASA consolidates and streamlines the total number of DRDs. The highest risk reduction and insight value DRD's pertain to requirements verification.	The DRDs allow for deliverables to be combined as practicable so long as the content is addressed. Intent is that content in DRDs is required in execution of the contract anyway. DRDs should not be a significant schedule driver and are expected to be reflected in the price of procurement.
20	DRFP - 80GSFC26R0011	Performance Incentives: The performance incentives include factors that are outside the Contractor's control, for example DWE Availability is based on DSN scheduling which is in NASA's control. Question: How will NASA factor in achievement of performance objectives that are outside of the Contractor's control?	Per Performance Incentive (Mission Operations) paragraph c, If it is determined that the Government is the sole cause of the Contractor's inability to meet a threshold requirement, the Contracting Officer will adjust the affected requirement or performance period for incentive purposes.
21	DRFP - 80GSFC26R0011	The DRFP includes FAR 52.217-8 (Option to Extend Services), which allows NASA to extend contract performance by up to six months beyond the final option period (CLIN 0008), yet the DRFP provides no guidance on whether this clause applies exclusively to operations CLINs, how it should be priced, or whether any extension costs must remain within the $700M statutory ceiling. Will the Final RFP clarify the scope of FAR 52.217-8, provide pricing guidance for the extension period, and confirm whether any associated costs fall within or outside the $700M appropriation under P.L. 119-21?	FAR 52.217-8 (Option to Extend Services) removed from final RFP release.
22	DRFP - 80GSFC26R0011 (A.6 Proposal Preparation – General Instructions)	FAR 52.232-31(c)(2), incorporated at page 73 of the DRFP, permits contract financing payments of up to 15 percent of the contract price in advance of any performance of work. However, the evaluation criteria at Section 52.212-2(a)(1)(2)(4), page 50 of the DRFP, state that "front-loaded distributions of payments demonstrate lower levels of confidence," suggesting that any early milestone payment — even one that is fully compliant with FAR 52.232-31 — may result in a less favorable evaluation. Will NASA clarify whether a milestone payment proposed at or below the 15 percent statutory advance threshold will be evaluated neutrally, or whether Offerors should expect a negative evaluation impact for any payment proposed prior to a significant hardware or results-oriented milestone event?	Each Offeror needs to make their own business decsion and provide a proposal based on their own business practices. Advanced payments will be evaluated neutrally.
23	DRFP - 80GSFC26R0011 (Enclosure 1 - Draft PWS)	Section 9.5 outlines four (4) Software DRD's. The Offeror requests NASA confirms these DRD are associated with Ground Systems software versus Spacecraft and/or Payloads.	The Draft PWS was modified to move statements relevant to flight software into WBS 6 (spacecraft) while retiaining ground system software relevant statements in WBS 9. The DRDs applicability is provided in each DRD consistent with the PWS.

24	Industry Day Slides	NASA confirmed at Industry Day that all contractor CLINs must fit within the $700M appropriated under P.L. 119-21, yet the DRFP does not clarify whether this ceiling applies solely to the contractor's total evaluated price or whether it also encompasses NASA internal costs such as Government Technical Assistance (GTA), Government Furnished Property (GFP), and NASA project office costs across GSFC, GRC, and JPL. Will the Final RFP confirm whether the $700M ceiling is contractor-facing only, and if NASA internal costs draw from the same appropriation?	On Industry Day, no one said "all CLINs must fit within the $700M appropriated under P.L. 119-21". What was said is that all CLINs must be priced.
25	DRFP - 80GSFC26R0011 (Cover Letter)	Will T&M task orders on the IDIQ be issued unilaterally? Does the offeror have the opportunity to no-bid a T&M task order?	Refer to NFS 1852-216-80. NASA retains the ability to unilatery issue IDIQ tasks. NASA requires vendors to bid T&M (engineering analysis) task orders as required under the scope of the contract.
26	DRFP - 80GSFC26R0011	(Pg. 47, A.8.2(b)(5)) This lists "option for increased quantity" does this mean the Government wants the Offeror to propose a price that will be used for the initial Launch, integration, and orbit insertion and that price will also be used for any additional launch, integration and orbital insertions? Does the government plan to have multiple of these over the life of the contract?	The MTN contract does not plan to procure multiple or additional launch campaigns over the life of the contract. The price and schedule incentive proposed for CLIN 3 are specifically for the Launch, Integration, and Orbit Insertion required to deploy the proposed MTN system.
27	Attachment B- Ground Rules Item 14	(Pg 9/Section 4) Can NASA provide clarification on why the particular proximity link frequency bands are selected?	The NASA proximity frequency bands meet the throughput and allocations as provided by the Space Frequency Coordination Group (SFCG) recommendations. SFCG is an international coordination forum where civil space agencies work together to manage and harmonize the radio-frequency spectrum used by space missions, including operations at Mars. The final version of the requirements does include the SFCG recommended Ku-band frequencies. NASA will coordinate with SFCG and National Telecommunications Information Administration (NTIA) for frequency authorization. Additional details can be found in the updated proximity link requirements.
28	Attachment B- Ground Rules Item 13	(Pg 9/Section 4) Can NASA provide clarification on the definition of scheduling? Is the requirement for Human-In-The-Loop?	Human in the loop is not required. This is a requirement to coordinate multiple users of the MTN system, similar to what is currently implemented on DSN/MRN.
29	MTN-REQ-049	Does the NASA science payload have any requirements for pointing or steering? If it does, please specify those requirements; if not, please add a sentence explicitly stating there are none.	The SMD payload will not drive pointing or steering requirements.
30	MTN-REQ-049	Is there any flexibility in the packaging dimensions of the payload?	Science payload packaging flexibility within the envelope exists. The payload will not exceed the envelope provided.
31	MTN-REQ-049	Will the payload always be turned on or does it have a duty cycle?	Assume steady state with 100% duty cycle. This steady state payload power consumption should be used for budgeting purposes.
32	MTN-REQ-031	Would NASA consider X-band vs. UHF?	No, UHF is required for legacy systems. All surface assets do not support bi-directonal X-band communications.

33	Attachment F - Draft DRDs	Recommend including a statement that all DRDs can be tailored and/or combined.	The Data Procurement Document states the following: Unless otherwise noted in this or other superseding documents, data products and documentation produced by the offeror through existing internal practices may be used to satisfy any number of data requirements described herein (to the extent practicable) granted that: 1. The supplier clearly communicates traceability to the data deliverables being satisfied by each submission and, 2. The NASA MTN reviewer concurs that the submission adequately satisfies the intent of required deliverable in question.
34	Attachment G- Draft GFE	Are Contractors expected to incorporate the Electra UHF Transceiver (EUT) to support Legacy Proximity Links? If so, is two units the maximum or minimum number of units available?	The contractor can utilize any transceiver that meets the UHF requirements. It is not required to use the Electra UHF transceiver. Two Grade II+ units are available as GFE if the contractor elects to use these.
35	DRFP - 80GSFC26R0011 (Enclosure 1 - Draft PWS)	Recommend reducing the level of detail in the draft PWS to allow contractors propose programmatic approaches that align with internal processes and management style. Recommend that tailoring to the currently drafted of the PWS be considered during the proposal and/or during execution. We believe the focus should be on the ultimate goals of MTN and the PWS should be limited to delivering a system which meets those goals.	The current Draft PWS (Enclosure 1) already provides the flexibility requested. Offerors are explicitly permitted to structure their proposed PWS at their own discretion (they are also not required to use the provided standard Level 2 Work Breakdown Structure). Furthermore, Offerors may tailor the mandatory statements by proposing language that 'meets the intent of' the drafted requirements, provided those changes are highlighted in their proposal. This allows Offerors ample room to align the PWS with their own internal processes and commercial management styles.
36	DRFP - 80GSFC26R0011 (Enclosure 1 - Draft PWS)	It is assumed that the "major non-conformance meetings" NASA desire a voting membership role in are exclusive to the MTN effort, not spaceflight hardware that has commonality with MTN. Is this assumption correct?	The Government confirms this assumption is correct; NASA's voting membership role applies exclusively to major non-conformance boards (MRBs, FRBs, ARBs) adjudicating hardware, software, or systems directly allocated to the MTN project. However, if a major non-conformance, anomaly, or mishap investigation occurs in the performance of activities not under this contract (such as on common or fleet hardware) that is relevant to the MTN design, production, or operations, the Contractor shall notify NASA and make all associated data, root cause analyses, and resulting reports available to the Government.
37	DRFP - 80GSFC26R0011 (Enclosure 1 - Draft PWS)	What role does NASA have in approving or providing launch commit criteria, launch constraints and abort procedures?	NASA typically writes the Launch Commit Criteria (LCC) for our science mission observatories. In the case of MTN, since the vendor will be responsible for launch vehicle and spacecraft, NASA will develop and include LCC related to the SMD payload, and will of course need to understand and approve all LCC related to the spacecraft and launch vehicle.
38	DRFP - 80GSFC26R0011 (Enclosure 1 - Draft PWS)	What role does NASA have in coordinating launch manifest requirements and launch window constraints with the launch provider?	It is the Contractor's responsibility to coordinate launch manifest requirements and launch window constraints with the launch provider.

39	DRFP - 80GSFC26R0011 (Enclosure 1 - Draft PWS)	What will NASA do with the identified safety-critical and mission-critical flight software?	NASA will use this information to maintain insight into the safety-critical and mission-critical functionality of the system.
40	DRFP - 80GSFC26R0011 (Enclosure 1 - Draft PWS)	What is intended by government oversight in key verification events and test activities? Is the government intending to explicitly approve if the contractor has verified the requirements via key verification events and test activities (oversight) or is the government asking for transparency in how the contractor is performing these activities (insight)?	The Government requires technical insight into key verification events and test activities. Deliniation between insight and oversight is expected to be cleary defined through the DRD-PM-08, Insight and Oversight Implementation Plan. The Government will provide oversight and approval of verification closure notices per DRD-SE-13, Verification Closure Notices and Validation Closure Notices/Requirements Verification Traceability Matrix.
41	DRFP - 80GSFC26R0011 (Enclosure 2 - Draft QASP)	(Pg 2-3, Oversight) Items in this section seem similar to Product Assurance Actions (PAAs) that NASA performs now for other contracts (process witnesses, audits, examinations, record reviews). Is this the same level of oversight being requested?	Similar levels of oversight are expected for critical items. Specific oversight activities will be defined through DRD-PM-08, Insight and Oversight Implementation Plan.
42	DRFP - 80GSFC26R0011 (Enclosure 2 - Draft QASP)	Is the expectation for a NASA resident officer to be located at the contractor facility full time during contract execution?	Yes, with specific residency details defined in the DRD-PM-08, Insight and Oversight Implementation Plan.
43		Please confirm that a GFX launch vehicle can be proposed (eliminating CLIN3).	Offerors are required to propose on CLIN 0003.
44		Please provide a date for when the draft science requirements will be available (eg: ops planning & pointing, etc).	The science payload will not drive pointing or orbit selection. Science planning requirements will be provided once the payload is selected. The payload will not exceed the SWaP envelope provided and are accomodated on a do-no-harm basis.
45		Please specify the degree of pointing required for the MTN scientific payload.	The science payload will not require pointing or steering.
46		Please provide a definition for what is considered full operational readiness at Mars.	The intent of "full operational readiness" is that spacecraft checkout and commissioning are complete at Mars and that the system is ready to begin nominal operations at the planned end-state orbit. The proximity links and DWE links have been commissioned.
47	DRFP - 80GSFC26R0011	Please confirm the Government will add FAR 52.217-7 (Option for Increased Quantity-Separately Priced Line Item) for thirty days notification ahead of exercising the option.	The Government intends to provide the Contractor an advance notice thirty days ahead of exercising the option, through routine status meetings with the CO/COR.